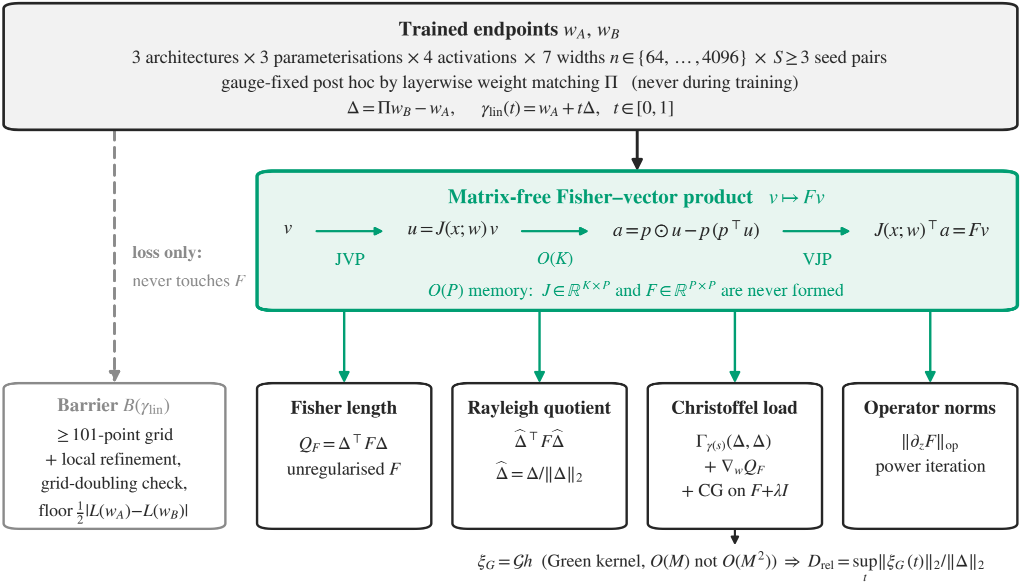

Not interchangeable: Hutchinson traces answer a typical-direction question, power iteration a worst-case one. A random probe returns $\mathbb{E}\, z^\top A z = \operatorname{tr} A / P = \|A\|_{\text{op}}\, r_{\text{eff}}(A)/P$, so with $NK \ll P$ it reports a flattening exponent that is too large.